

Manifold-Informed Architecture Benchmark

ALZHEIMERS: 2 classes, 8D input

Generated: 2026-04-15 09:41:10 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009) | 2026-04-14 22:20:05
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Best: Standard (64→32) test acc = 0.8170 ± 0.0337

Manifold: 8D → 6D (25.0% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	ALZHEIMERS
Input dimensionality	8D
Classes	2
Intrinsic dim (d)	6
Variance threshold (τ)	0.9
Epochs	150
Trials	3
Batch size	32
Learning rate	0.001
k neighbors (local PCA)	20

Manifold Discovery

Local PCA over the training set, $k = 20$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 2$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	5.3	0.5	4	6	34.2
0.90	4.6	0.5	4	6	42.6
0.85	4.0	0.4	3	5	49.5
0.80	3.6	0.5	3	5	54.9

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
1	5.0	0.3	4	6
0	4.2	0.4	4	5

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.7532 ± 0.0289	—	—
ManifoldModel ($\tau=0.9$)	0	0.7404 ± 0.0434	—	—
	2,722	0.8170 ± 0.0337	0.6774	0.3002
Standard (64→32) ★				
Manifold (2d→d, d=6)	68	0.8014 ± 0.0558	0.4074	11.7856
Wide Manifold (d+1=7)	79	0.7929 ± 0.0518	0.4058	10.0368
Manifold+ManifoldAdam (d=6)	68	0.7929 ± 0.0422	0.4199	11.6604
PCA→6D + MLP (2d→d)	176	0.8099 ± 0.0325	0.4157	4.6019
Intrinsic Dim (PCA→6D→C)	56	0.8170 ± 0.0520	0.3789	14.5897

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (64→32) — test accuracy 0.8170 ± 0.0337
- **Manifold compression:** 8D → 6D (25.0% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009)
Commit date	2026-04-14 22:20:05 -0400
Commit message	chore(release): bump version to 0.6.0
Python	3.12.13
TensorFlow	2.21.0
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-04-15 09:41:10

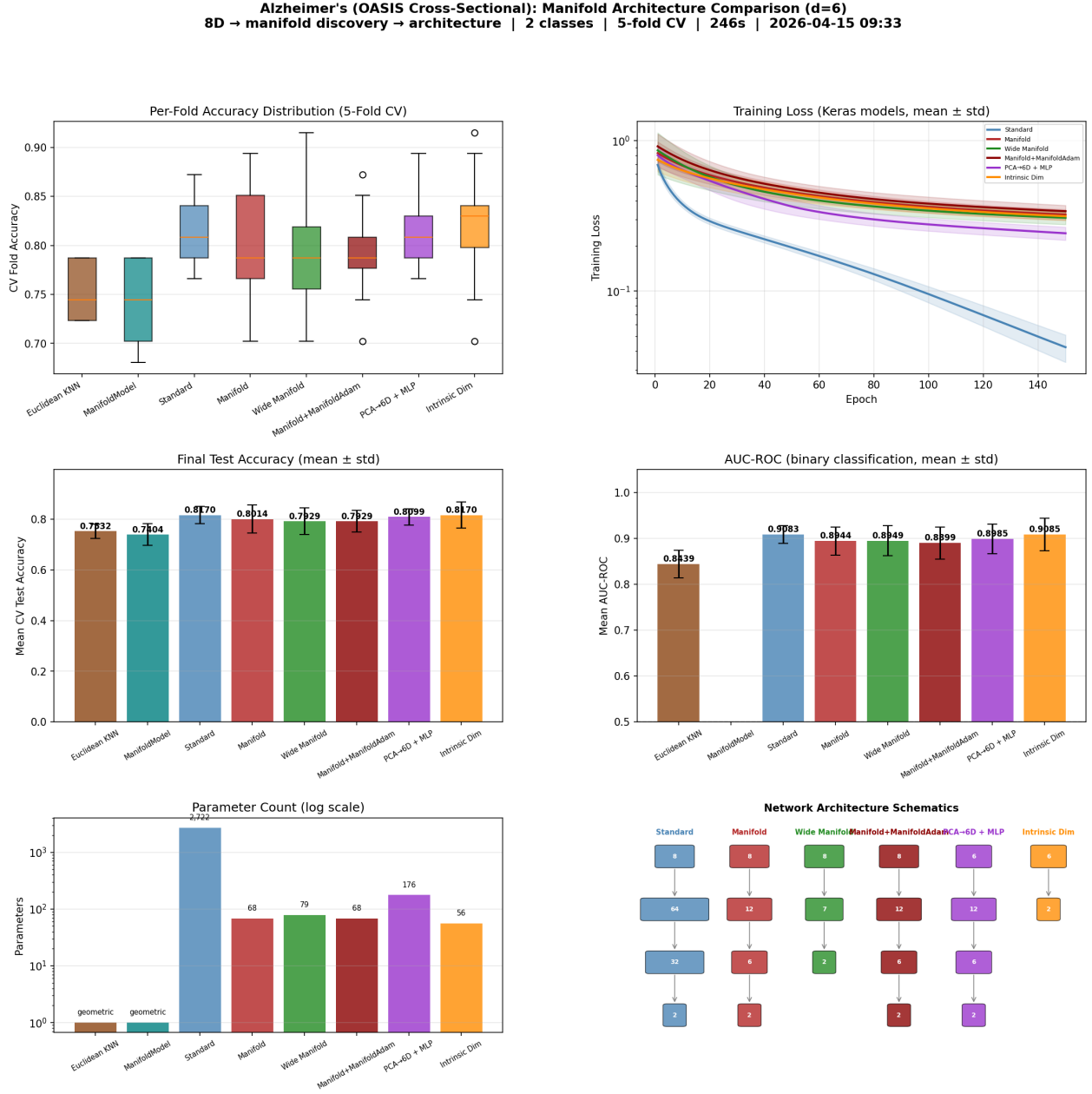


Figure 1: ALZHEIMERS manifold-informed vs. standard architecture comparison. $d = 6$, 150 epochs, 3 trials.